

Maqsurdur Rahman

Publication List

Ph.D. Candidate in Computer Science, Boise State University, USA (Expected May 2027)

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Google Scholar: Google Scholar Profile
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Publication Summary

My publication record includes **26 peer-reviewed or accepted scholarly works**, including **4 Ph.D.-stage peer-reviewed or accepted publications** in robust quantum machine learning, trustworthy AI, medical image analysis, and hybrid quantum-classical systems. In addition, I have **1 manuscript currently under review at NeurIPS 2026**. My earlier publications include peer-reviewed journal articles, international conference papers, and a book chapter in machine learning, cybersecurity, NLP, data mining, intelligent systems, and network/security applications.

Research alignment: Artificial Intelligence; Secure and Trustworthy Machine Learning; Information Security; High-Performance Computing for AI; Bioinformatics/Clinical AI; Medical Image Analysis; Information Visualization; Software and AI Systems.

Manuscript Under Review

1. **M. Rahman** and J. Zhuang, “**HQADNet: Contrastive Prototype-Supervised Quantum Representation Learning for Robust Medical Image Classification against Backdoor Poisoning.**” *Manuscript under review, NeurIPS 2026*.

Selected Peer-Reviewed and Accepted Ph.D.-Stage Research Publications

1. **M. Rahman** and J. Zhuang, “**QAPNet: A Quantum-Attentive Patchwise Network for Robust Medical Image Classification Under Noisy Inputs.**” *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2026. **Accepted for presentation at AAAI-26**.
2. **M. Rahman** and J. Zhuang, “**NQNN: Noise-aware Quantum Neural Networks for Medical Image Classification.**” *Medical Image Computing and Computer-Assisted Intervention (MICCAI) 2025*, LNCS, Springer. Accepted/published conference paper.
3. **M. Rahman**, R. Liu, A. Majumder, P. C. Paul, K. Mo, A. Begum, K. Sultana, N. Akter, L. Wei, Y. Zhang, and J. Zhuang, “**Benchmarking Barren Plateau Mitigation Strategies in Quantum Neural Networks on Standard and Medical Image Datasets.**” *Journal of Imaging*, vol. 12, article 275, 2026. DOI: 10.3390/jimaging12070275.
4. **M. Rahman**, P. C. Paul, A. Begum, K. Sultana, N. Akter, A. Majumder, M. Zhu, Z. Sheng, W. Xin, X. Jin, and J. Zhuang, “**Systems-Level Support for Hybrid Quantum-Classical Learning: A Systematic Review with a Medical Imaging Translation Lens.**” *Journal of Imaging*, vol. 12, article 232, 2026. DOI: 10.3390/jimaging12060232.

Earlier Peer-Reviewed Journal Articles

1. **M. Rahman**, M. T. Ahmed, S. Nur, and A. Z. M. Islam, “The Prediction of COVID-19 Pandemic Outbreak on Bangladesh Perspective Using Multiple Machine Learning Algorithms: A Comparative Study.” *International Journal of Electrical and Computer Engineering (IJECE)*, 2022.
2. M. T. Ahmed, **M. Rahman**, S. Nur, A. Z. M. Islam, and D. Das, “NLP and Machine Learning Based Cyberbullying Detection for Bangla and Romanized Bangla Texts.” *TELKOMNIKA (Telecommunication Computing Electronics and Control)*.
3. K. Farhana, **M. Rahman**, and M. T. Ahmed, “An Intrusion Detection System for Packet and Flow Based Networks using Deep Neural Network Approach.” *International Journal of Electrical and Computer Engineering (IJECE)*.

4. P. C. Paul, **M. Rahman**, A. Begum, M. T. Ahmed, D. Chakraborty, and M. S. Rahman, "Combining BERT with LDA: Improved Topic Modeling in Bengali Language." *IAENG International Journal of Computer Science*, vol. 52, no. 2, pp. 383–393, 2025.
5. **M. Rahman** and M. Shahjahan, "Exploring the Taxonomy of Survey Papers on Large Language Models Using Classical Machine Learning." *Journal of Intelligent Learning Systems and Applications*, vol. 17, no. 2, pp. 108–122, May 2025. DOI: 10.4236/jilsa.2025.172006.
6. A. H. Antar, M. T. Ahmed, M. Akter, M. S. Rahman, **M. Rahman**, P. C. Paul, and N. Parvin, "Deep Neural Network Based Spam Email Classification Using Attention Mechanisms." *Journal of Intelligent Learning Systems and Applications*, vol. 15, pp. 144–164, 2023.
7. M. T. Ahmed, N. Akter, **M. Rahman**, D. Das, T. A. Z. M., and G. Rashed, "Multimodal Cyberbullying Meme Detection from Social Media Using Deep Learning Approach." *International Journal of Computer Science and Information Technology (IJCSIT)*, vol. 15, pp. 27–37, 2023.
8. M. T. Ahmed, A. H. Antar, **M. Rahman**, A. Z. M. T. Islam, D. Das, and M. G. Rashed, "Social Media Cyberbullying Detection on Political Violence from Bangla Texts Using Machine Learning Algorithm." *Journal of Intelligent Learning Systems and Applications*, vol. 15, no. 4, pp. 108–122, 2023.
9. M. T. Ahmed, A. A. Rubi, S. Rahman, and **M. Rahman**, "Red-AODV: A Prevention Model of Black Hole Attack for VANET Protocols and Identification of Malicious Nodes in VANET." *International Journal of Computer Networks and Applications (IJCNA)*. DOI: 10.22247/ijcna/2021/209985.

Peer-Reviewed Conference Papers

1. **M. Rahman**, S. Nur, M. T. Ahmed, D. Das, and A. Z. M. Islam, "A Feature Engineering Approach for Detecting Cyberbullying in Bangla Text Using Machine Learning." *2022 International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET)*.
2. M. T. Ahmed, **M. Rahman**, S. Nur, A. Z. M. Islam, and D. Das, "Introduction of PMI-SO Integrated with Predictive and Lexicon Based Features to Detect Cyberbullying in Bangla Text Using Machine Learning." *2nd International Conference on Artificial Intelligence: Advances and Applications (ICAIAA 2021)*.
3. M. T. Ahmed, **M. Rahman**, S. Nur, A. Z. M. Islam, and D. Das, "Deployment of Machine Learning and Deep Learning Algorithms in Detecting Cyberbullying in Bangla and Romanized Bangla Text: A Comparative Study." *IEEE International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT 2021)*.
4. A. S. Urmi, M. T. Ahmed, **M. Rahman**, and A. Z. M. Islam, "A Proposal of Systematic SMS Spam Detection Model Using Supervised Machine Learning Classifiers." *International Conference on Computer Vision and Robotics (CVR 2021)*.
5. S. Palit, S. Nur, Z. Khatun, **M. Rahman**, and M. T. Ahmed, "Analysis of Online Education System of Bangladesh during COVID-19 Pandemic Based on NLP and Machine Learning: Problem and Prospect." *First International Conference on Emerging Trends in Industry 4.0 (ETI 4.0, 2021)*.
6. M. Islam, **M. Rahman**, M. T. Ahmed, A. Z. M. Islam, D. Das, and M. M. Hoque, "Sexual Harassment Detection using Machine Learning and Deep Learning Techniques for Bangla Text." *International Conference on Electrical, Computer and Communication Engineering (ECCE 2023)*.
7. P. C. Paul, M. S. Uddin, M. T. Ahmed, M. M. Hoque, and **M. Rahman**, "Semantic Topic Extraction from Bangla News Corpus Using LDA and BERT-LDA." *25th International Conference on Computer and Information Technology (ICCIT 2022)*.
8. I. Hossain, **M. Rahman**, T. Ahmed, and A. Z. M. Islam, "Forecasting the Rating of Online Products from Customer Text Review based on Machine Learning Algorithms." *International Conference on Information and Communication Technology for Sustainable Development (ICICT4SD)*.

9. I. Hossain, **M. Rahman**, T. Ahmed, A. Z. M. Islam, and S. Rahman, “Rating Prediction of Product Reviews of Bangla Language Using Machine Learning Algorithms.” *International Conference on Artificial Intelligence and Mechatronics Systems (AIMS 2021)*.
10. P. Barman, **M. Rahman**, and M. T. Ahmed, “A Comparison of Statistical and Deep Learning Model in Forecasting Time Series.” *International Conference on Sustainable Development in Technology for 4th Industrial Revolution (ICSDTIR)*.
11. P. Shil, U. S. Rahman, **M. Rahman**, and M. S. Islam, “An Approach for Detecting Bangla Spam Comments on Facebook.” *International Conference on Electronics, Communications and Information Technology (ICECIT 2021)*.
12. M. Rahman, M. S. Rahman, N. Parvin, and **M. Rahman**, “Fundamental Frequency Extraction by Utilizing the Modified Weighted Autocorrelation Function in Noisy Speech.” In: *International Conference on Advances in Data-driven Computing and Intelligent Systems*, pp. 65–76, Springer, 2023.

Book Chapter

1. **M. Rahman** and S. Hossain, “Performance Evaluation of Social-Aware Routing Protocols in an Opportunistic Network.” In: *Opportunistic Networks: Mobility Models, Protocols, Security & Privacy*.